DPP- 02

CS & IT

Algorithm

Analysis of Algorithm

Q1 Which of the following natation is/are transitive but not reflexive

(A) Big oh (O)

(B) Big omega (Ω)

(C) Small oh (o)

(D) Small omega (ω)

Q2 If f(n) =$\sum_{i=1}^{n} i^3$

Then which of the following choices is/are true for f(n)?

(A) $\theta\left(n^4\right)$ (B) $\Omega\left(n^4\right)$

(C) $\theta\left(n^5\right)$ (D) $\Omega\left(n^3\right)$

Q3 Consider the following program:

```
  main ( )
{
 P = n!;
 for (i = 1; i <= n ; ++i)
     for (j = 1 ; j <= P ; 2*j )
 C = C + 1;
 }
```

What is the time complexity of above code?

(A) $O\left(n^2\right)$ (B) $O\left(n^2 \log n\right)$

(C) O (nlogn) (D) O(n)

Q4 Consider the following code:

```
main ( )
{
i = 1, j = 1;
while (j <= n )
{
  ++ i;
  j  = j + i;
}
```

What is the time complexity of above code?

(A) $\theta(n)$

(B) $\theta(\sqrt{n})$

(C) $\theta(\log n)$

(D) $\theta(n\log(\log n))$

Q5 Consider the following code:

```
Algorithm T(n)
{
if (n = =1) return;
else
{
T(n/2);
}
}
```

What is the space complexity of above code?

(A) $\theta(\log n)$ (B) $\theta(n)$

(C) $\theta(n\log\log n)$ (D) $\theta(\sqrt{n})$

Q6 $f(n)= 2^{n^2}$, $g(n)= n!$, $h(n)= 2^{\log n^2}$

Which of the following is/are correct?

(A) f(n) =Ω(g(n))

(B) h(n) =Ω(g(n))

(C) h(n) =O(g(n))

(D) g(n)=Ω(f(n))

Q7 Consider the following rotations:

1. $\sqrt{\log n} = O(\log\log n)$
2. $\log n = \Omega\left(\frac{1}{n}\right)$
3. $n^2 = \theta\left(2^{2\log n}\right)$
4. $(0.061)^n = \theta(1.02)^n$

How many notations is/are correct?______________.

Q8 Consider the following functions:

$$f_1 = 2^n$$
$$f_2 = n!$$
$$f_3 = n^n$$
$$f_4 = e^n$$

What is the correct increasing order of above function?

(A) f1 f4 f2 f3

(B) f2 f1 f4 f3

(C) f2 f4 f1 f3

(D) f2 f1 f4 f3

Answer Key

Q1	C, D
Q2	A, B, D
Q3	B
Q4	B
Q5	A
Q6	A, C
Q7	2
Q8	A

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

	O	Ω	θ	o	ω
Reflective				x	x
Transitive					

Small oh (o) and small omega(ω) comes under transitive but not reflective.

Q2 Text Solution:

f(n) = $\sum_{i=1}^{n} x^3$

i = 1

= $1^3 + 2^3 + 3^3 + 4^3 + 5^3 + \ldots n^3$

= $\left[\frac{n(n+1)}{2}\right]^2$

= O(n^4), θ(n^4), Ω(n^4)

A, B, D options are correct.

Q3 Text Solution:

Time complexity = n * log p

$\downarrow$ $\downarrow$

first loop second loop

= n* log n!

= n * nlogn ,[logn!=θnlogn]

= $O\left(n^2 \ \log n\right)$

Q4 Text Solution:

i = 1, 2 , 3................................x

k =(1+2) (1+2) (1+2+3) (1+2+3+...x)

Time complexity = 1 + 2 + 3 + 4 +.......x $\leq$n

$$\Rightarrow \frac{x(x+1)}{2} \leq n$$

$$\Rightarrow x^2 + x = 2n$$

$$\Rightarrow x^2 \cong 2n$$

$$\Rightarrow x = \sqrt{n}$$

$$\Rightarrow x = \theta(\sqrt{n})$$

Q5 Text Solution:

Time complxity = $T\left(\frac{n}{2}\right)+1$

$= \theta(\log n)$

Space complixity = we are pushing k activation record for n = 2^k

$n = 2^k$

$\boxed{k = (\log_2 n)}$

Space = $\boxed{\theta(\log_2 n)}$

Q6 Text Solution:

$f(n) = 2^{n^2}$

$g(n) = n!$

$h(n) = 2^{\log_2 \ n^2}$

$= \left(n^2\right)^{\log_2 \ 2}$

= n^2

- 2^{n^2} $\quad$ n!

taking both side log

$\log 2^{n^2}$ $\quad$ logn!

n^2 $\quad$ nlogn

$n^2 = \Omega(n\log n)$

$f(n) = \Omega(g(n))$

- $h(n) = \Omega(f(n))$

$n^2 \leq n!(\text{True})$

- $g(n) = \Omega(f(n))$

$g(n) \geq 2^{n^2}$

$n! \geq 2^{n^2}$ (False)

Q7 Text Solution:

1. $\sqrt{\log n} = 0(\log\log n)$

$(\log n)^{\frac{1}{2}} \leq \log\log n$

$\frac{1}{2}\log\log n \leq \log(\log\log n)$ (False)

2. $\log n = \Omega\left(\frac{1}{n}\right)$

$\log n \geq \frac{1}{n}$ (True)

3. $n^2 = \theta\left(2^{2\log n}\right)$

$n^2 = 2^{2\log n}$

$n^2 = 2^{\log n^2}$

$n^2 = \left(n^2\right)^{\log_2^2}$

$n^2 = n^2(\text{True})$

4. $(0.0161)^n = \theta(1.02)^n$ (False)

Q8 Text Solution:

- $2^n < e^n$
- $n! < n^n$
- $(2^n, e^n) < n!$
- $2^n < e^n < n! < n^n$

$f_1 < f_4 < f_2 < f_3$